

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. Examination (CE/IT/EC)

May 2013

CL102.01 Environmental Science

Date: 09.05.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1(a) Discuss the relationship between man and environment. [03]
(b) Explain the impact of technology on the environment. [04]
- Q - 2(a) Define noise. Briefly describe the sources and effects of noise pollution. [04]
(b) Discuss the role of advanced technology in the degradation of environment. [05]
(c) Explain the objectives and guiding principles of Environmental Studies. [05]

OR

- Q - 2(a) Explain the effects of air pollutants on human health. [04]
(b) Give classification of water pollutants and describe organic water pollutants. [05]
(c) Explain structure of atmosphere with neat sketch in detail. [05]
- Q - 3(a) Explain general classification of air pollutants and describe classification according to sources. [04]
(b) Write short note on Eutrophication. [05]
(c) Explain Green House Effect. What are various sources of green house gases? [05]

OR

- Q - 3(a) Enlist different air pollutant with their sources and effects. [04]
(b) Define water pollution and enlist various sources of water pollution. Explain any one in detail. [05]
(c) What is 'Acid Rain'? Write a brief note on effects of acid rain on environment. [05]

SECTION – II

- Q - 4(a)** What are the Components of Environment? Explain in detail Lithosphere. [05]
- (b)** Answer any two questions. [02]
- (i) According to ecological food chain, man is a
(a) Decomposer (b) Producer (c) Consumer (d) None of these.
- (ii) Biotic Factor of eco-system is
(a) Sunlight (b) Soil (c) Consumer & producer (d) Wind
- (iii) Transpiration means water removal from
(a) Ocean (b) Plants (c) Ponds (d) Lakes

- Q - 5(a)** Define the terms: [04]
- (i) Autecology (ii) Synecology (iii) Photoautotroph (iv) Chemoautotroph.
- (b)** Describe briefly: [04]
- (i) Food chain. (ii) Food web.
- (c)** Describe components of Eco-system in detail. [06]

OR

- Q - 5(a)** What are the objectives of ecological studies? [04]
- (b)** 'The length of food chain is restricted to 4-5 levels' – Explain in detail. [04]
- (c)** What is meant by ecological pyramid? Explain the different types of pyramids giving suitable examples. [06]
- Q - 6(a)** Explain in detail conventional and non-conventional energy sources. [07]
- (b)** Discuss carbon cycle with neat sketch. [04]
- (c)** Describe major aspects of Eco-system. Also explain Desert Eco-system. [03]

OR

- Q - 6(a)** Explain Destruction versus Conservation of Energy Resources. [07]
- (b)** How Energy flows in the eco-system? Explain in detail. [04]
- (c)** Answer any three questions: [03]
- (i) Earthworms & bacteria are called _____.
- (ii) Explain the term 'community'.
- (iii) Write biochemical formula which describes the process of photosynthesis.
- (iv) Differentiate Macronutrients & Micronutrients.
